


January 8, 2026

Base 2, 8, 16

$$\begin{array}{cccccccccccc} & 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 & & & & & \\ \text{Binary } (11111111) & \rightarrow & (& & & & & &) & & & & & \\ & & & & & & & & & & & & & \end{array}$$
$$1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$
$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 =$$
$$= 255$$

$$(377)_8 = 3 \cdot 8^2 + 7 \cdot 8^1 + 7 \cdot 8^0 = 255$$

$$(FF)_{16} \rightarrow ()_{10}$$

base 16:

$$16^{\overset{1}{F}} + 16^{\circ F} = 16 \cdot 15 + 15 = 255_{10}$$

Decimal $()_{10}$ $\rightarrow$ $()_2$
 $\rightarrow$ $()_8$
 $\rightarrow$ $()_{16}$

10
11
12
13
14
15
A
B
C
D
E
F

HEX

$$\begin{array}{r}
 255 \mid 2 \\
 \hline
 254 \mid 127 \mid 2 \\
 \hline
 126 \mid 63 \mid 2 \\
 \hline
 62 \mid 31 \mid 2 \\
 \hline
 30 \mid 15 \mid 2 \\
 \hline
 14 \mid 7 \mid 2 \\
 \hline
 6 \mid 3 \mid 2 \\
 \hline
 2 \mid 1 \\
 \hline
 \end{array}$$

$255_{10} = 1111111_2$
 (A blue arrow points from the bottom of the binary representation to the first remainder '1' in the division process.)

114₁₀

$$\begin{array}{r|l} 114 & 2 \\ \hline 114 & 57 \\ \hline 0 & 56 \\ & 28 \\ & 14 \\ & 7 \\ & 3 \\ & 1 \end{array}$$

1110010₂

$$(0.12)_{10} \rightarrow (\quad)_2 = (.000111)_2$$

$0.12 \times 2 = 0.24$	0	just fraction ↓
$0.24 \times 2 = 0.48$	0	
$0.48 \times 2 = 0.96$	0	
$0.96 \times 2 = 1.92$	1	
$0.92 \times 2 = 1.84$	1	
$0.84 \times 2 = 1.68$	1	

Dec.	Binary	Dec.	Binary
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111
8	1000		

A B C D E F
Hex.

Binary addition

$$\begin{array}{r}
 \times \quad 01100 \\
 \quad 10001 \\
 \hline
 \quad 11101
 \end{array}$$

$$\begin{array}{r}
 \overset{1}{+} \overset{1}{1} \overset{1}{0} \overset{1}{1} 1 0 \\
 1 0 1 1 1 \\
 \hline
 \underline{1} [0 1 1 0 1]
 \end{array}$$

$$(1)_2 + (1)_2 = (10)_2$$

$$(\overset{1}{1}0)_2 + (1)_2 = (11)_2$$

It is overflow, machine should know that it is overflow

Binary subtraction

$$\begin{array}{r}
 \overset{0}{-} \overset{0}{0} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{1}{1} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{0}{0} \\
 \hline
 1
 \end{array}
 \quad
 \begin{array}{r}
 \overset{0}{-} \overset{1}{1} \\
 \hline
 1
 \end{array}$$

$$\begin{array}{r}
 - 10110 \\
 10010 \\
 \hline
 00100
 \end{array}$$

$$\begin{array}{r}
 - 10110 \\
 10011 \\
 \hline
 00011
 \end{array}$$

1's complement

logic operation

$$(10101111)_2$$

1's complement is inverse of digits

$$(01010000)_2 \text{ 1's comp.}$$

$$(01010001)_2 \text{ 2's complement}$$

$$(0101)_2 = (5)_{10}$$

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$(r-1)$'s r 's complements

radix = base

$$r=10 \quad L=4 \quad N=2468 \quad r^4=10000$$

$$r^4 - 1 = 9999$$

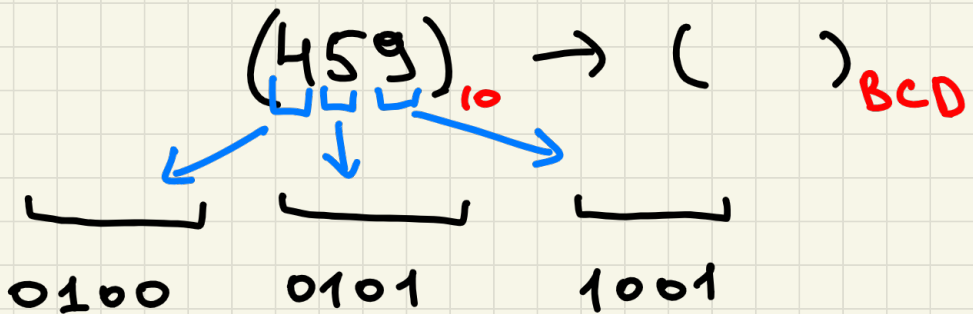
Therefore

$$\begin{array}{r} 9999 \\ - 2468 \\ \hline \underline{7531} \end{array}$$

Decimal $(185)_{10}$

Binary $(10111001)_2$

Binary Coded Decimal BCD



$$4_{10} + 8_{10} = 12_{10}$$

BCD range
(0-9)

$$0100_2 + 1000_2 = 1100_2 \rightarrow 12_{10}$$

$$0110_2 \rightarrow 6_{10}$$

$(\underline{0001} \underline{0010})_{BCD}$

$$\begin{array}{r}
 + 162 \\
 769 \\
 \hline
 931
 \end{array}$$

Binary Coded Decimal

$$\begin{array}{r}
 0001 \quad 0110 \quad 0010 \\
 0111 \quad 0110 \quad 1001 \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 1000 \quad \underline{1100} \quad \underline{1011} \\
 \text{doesn't exist in BCD} \quad \text{more than 9} \\
 \text{to solve add } 6_{10}
 \end{array}$$

$$\begin{array}{r}
 0110 \quad 0110 \\
 \hline
 \underline{1001} \quad \underline{0011} \quad \underline{0001} \\
 9 \quad 3 \quad 1
 \end{array}$$

encoding $\rightarrow$ public UTF-8
 encryption $\rightarrow$ special key

BOOLEAN LOGIC

$$A+B \rightarrow A \text{ or } B$$

$$A.B \rightarrow A \text{ and } B$$

$$A', \bar{A} \rightarrow \text{NOT } A$$

inputs

A	B	A.B	A+B	A'	B'
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0

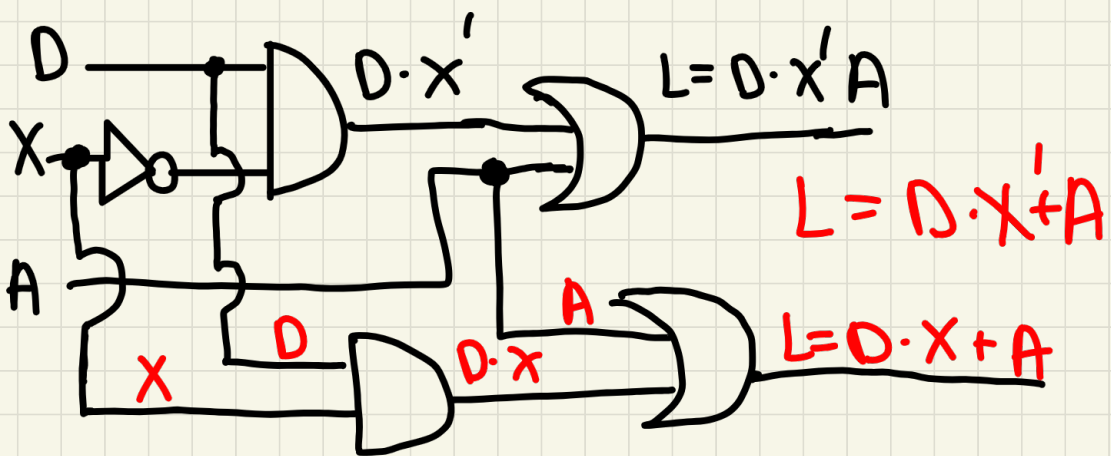
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$$L = \underbrace{D \cdot X'}_{\substack{\uparrow \\ \text{output}}} + \underbrace{A}_{\substack{\uparrow \\ \text{inputs}}}$$

D	X	A	X'	D · X'	D · X' + A
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0

output

0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1



A	B	C	AB	A'C	BC		AB + A'C
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0

1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

$$F(A, B, C) = (A+B)(A+C)$$

$$AA + AC + AB + BC \quad \text{Distributivity}$$

$$A + AC + AB + BC \quad A = AA$$

$$A(\underbrace{1 + C + B}_1) + BC$$

$$A \cdot 1 + BC = A + BC$$

Sum of Products SOP

$$F(A, B, C) = A'B'C + ABC + A'BC'$$

Product of Sum

$$\begin{aligned}
 F(A,B,C) &= (A+B)(A+C) \\
 &= (A+A) + A+C + AB+BC \\
 &= A + A + C + AB+BC \\
 &= A(1+C) + BC \\
 &= A \cdot 1 + BC = A+BC
 \end{aligned}$$

$$\begin{aligned}
 F &= A + BC + EF \\
 &\text{Sum of Product terms} \\
 G &= (A+B)(C+D+E)(F+H) \\
 &\text{Product of Sum terms}
 \end{aligned}$$

(k)	D	X	A	X'	D'X	D'X'+A		
0	0	0	0	1	0	0	←	(D'X'A)
1	0	0	1	0	0	1		
2	0	0	1	0	0	0		
3	0	1	0	1	0	1	←	D'XA
4	1	0	0	1	1	1	←	D'X'A'
5	1	0	1	0	1	1	←	D'X'A
6	1	1	0	0	0	0		
7	1	1	1	0	0	1	←	D'XA

January 20, 2026

Q 1

$$y + x'z + xy' = x + y + z$$

$$y + xy' + x'z$$

$$(y+x)(y+y') + x'z$$

$$y + x + x'z$$

$$y + (x+x')(x+z)$$

$$y + x + z = x + y + z$$

Q₂

$$x'y' + x'y + xy = x'y$$

$$x'(y' + y) + xy$$

$$x' + xy$$

$$(x' + x)(x' + y)$$

$$1 \cdot x' + y = x'y$$

Q₃

$$x'y' + y'z + xz + xy + yz' \quad 13 \text{ operations}$$

$$x'y' + y'z(x + x') + xz + xy + yz'$$

$$\underline{x'y'} + xy'z + \underline{x'y'}z + xz + xy + yz'$$

$$x'y'(1 + z) + xy'z + xz + xy + yz'$$

$$x'y' + \underline{xy'z} + \underline{xz} + xy + yz'$$

$$x'y' + xz(y' + 1) + xy + yz'$$

$$x'y' + xz + xy + yz'$$

$$x'y' + xz + xy(z+z') + yz'$$

$$x'y' + xz + xyz + xyz' + yz'$$

$$x'y' + xz + xyz + yz'(x+1)$$

$$x'y' + \underline{xz} + \underline{xyz} + yz'$$

$$x'y' + xz(1+z) + yz'$$

$$x'y' + xz + yz'$$

8 operations

SOP

$$F(a,b,c) = a + bc + abc'$$

$$A(a,b,c) = \underbrace{abc}_{\text{Minterms}} + \underbrace{a'bc}_{\text{Minterms}} + \underbrace{abc'}_{\text{Minterms}} \quad \text{Canonical (product terms)}$$

K						Minterms
	a	b	c	A	A	SOP
0	0	0	0	$a'b'c'$	0	$x=1$ $x'=0$
1	0	0	1	$a'b'c$	0	
2	0	1	0	$a'bc'$	0	
3	0	1	1	$a'bc$	1	
4	1	0	0	$ab'c'$	0	
5	1	0	1	$ab'c$	0	
6	1	1	0	abc'	1	
7	1	1	1	abc	1	

$$A(a,b,c) = \sum m(3,6,7) \rightarrow \text{minterms}$$

↓
sum of products

POS (product of sum)

$$G(a,b,c) = \underline{(a+b+c')} \cdot \underline{(a'b'+c')} \cdot \underline{(a+b+c)}$$

Max terms:

$$x=0$$

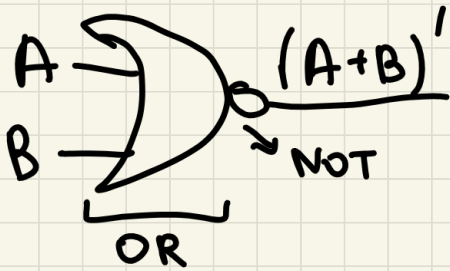
POS

$$x'=1$$

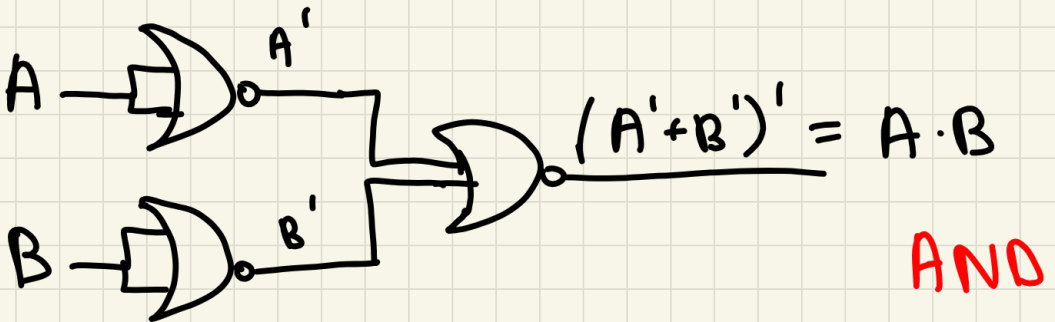
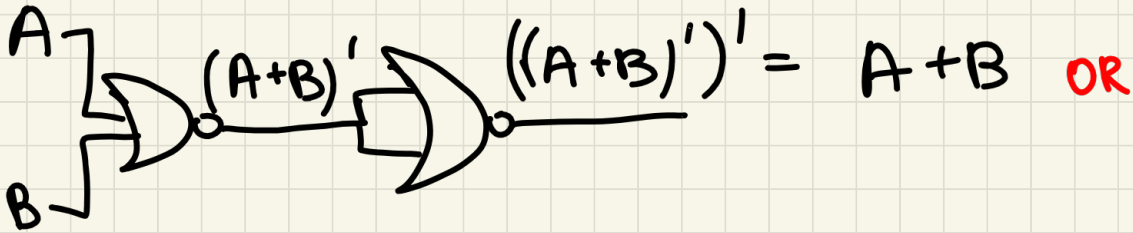
	a	b	c	P	
0	0	0	0	abc	1
1	0	0	1	abc'	1
2	0	1	0	$ab'c$	0
3	0	1	1	$ab'c'$	0
4	1	0	0	$a'b c$	0
5	1	0	1	$a'b c'$	0
6	1	1	0	$a'b'c$	0
7	1	1	1	$a'b'c'$	1

$$P(a,b,c) = \underbrace{\Pi}_{\text{POS}} \underbrace{M}_{\text{max terms}} (0, 1, 7)$$

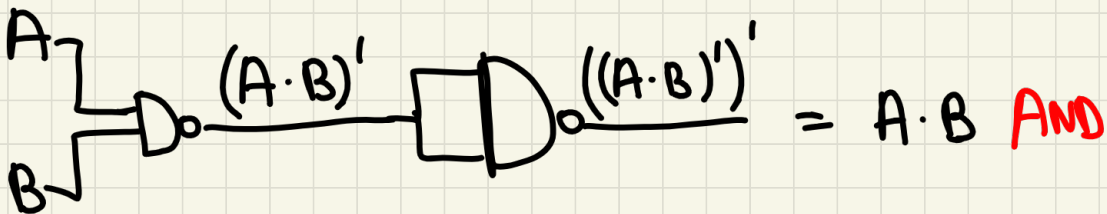
Always write Maxterms POS $x'=1$
Minterms SOP $x=1$
 $x=0$
 $x'=0$



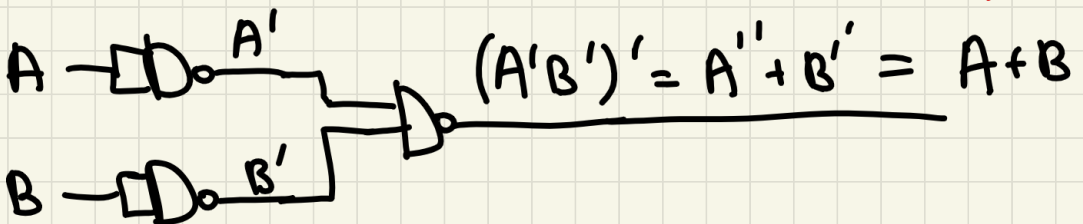
NOR



AND



OR



$$F(a, b, c) = a'b'c + bc' + ac' \text{ (Standard)}$$

↓
missing a
↓
missing b

$$F(a, b, c) = a'b'c + \cancel{abc'} + a'bc' + \cancel{abc'} + \underline{ab'c'}$$

$$F(a, b, c) = a'b'c + a'bc' + abc' + ab'c'$$

SOP Canonical form

$$F(a, b, c) = (a+b+c) \cdot (a+b) \cdot (a+c') \text{ (Standard)}$$

↓
needs c
↓
needs B

$$\cancel{(a+b+c)} \cdot \underline{(a+b+c)} \cdot \cancel{(a+b+c')} \cdot \underline{(a+b+c')} \cdot (a+b'+c')$$

$$(a+b+c) (a+b+c') (a+b'+c')$$

POS Canonical form

1-Variable map

x	F(x)
0	F(0)
1	F(1)

2 variable maps

x	y	F(x)
0	0	F(0)
0	1	F(1)
1	0	F(2)
1	1	F(3)

	0	1
0	00	01
1	10	11

2 variable maps

x	y	z	F(x)
0	0	0	F(0)
0	0	1	F(1)
0	1	0	F(2)
0	1	1	F(3)
1	0	0	F(4)
1	0	1	F(5)
1	1	0	F(6)
1	1	1	F(7)

x \ yz	00	01	11	10
0	$x'z'y'$ 000 0	$x'y'z$ 001 1	$x'yz$ 011 3	$x'yz'$ 010 2
1	100 4	101 5	111 7	110 6

→ gray codes

SOP | $x=1$
 $x'=0$

$$F(x,y,z) = \sum m(1,3,5,7)$$

$$x'y'(z+z')$$

$$x'y'(1) = x'y'$$

0	0 ₀	1 ₁	1 ₃	0 ₂
1	0 ₄	1 ₅	1 ₇	0 ₆

Group of 4

$$(2^n) = 1, 2, 4, 8, 16$$

$$x' y' z + x' y z + x y z' + x y z$$

$$z (x' y' + x' y + x y' + x y)$$

$$z (x' \underbrace{(y' + y)}_1 + x \underbrace{(y' + y)}_1)$$

$$z (\underbrace{x' + x}_1)$$

z

Shorter way to calculate

4 variable map

w x y z | f(w, x, y, z)

16 rows

w \ x \ yz	00	01	11	10
00	0000 0	0001 1	0011 3	0010 2
01	0100 4	0101 5	0111 7	0110 6
11	1100 12	1101 13	1111 15	1110 14
10	1000 8	1001 9	1011 11	1010 10